# 棋馆的最后一局

一个老师、一个学生和一台不会累的机器

## The Last Game at the Go Club

*An old teacher, his best student and a machine that never gets tired*

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Footnotes mark names, festival terms and words above HSK 6 (HSK 3.0 lists: github.com/ivankra/hsk30, MIT).
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1 杭州[1]有很多老巷子[2]。离西湖[3]不远的一条小巷子里，有一家很小的棋馆[4]，门口挂着一块旧木牌，上面写着四个字：青石棋馆[5]。棋馆开了三十多年，主人姓钱，大家都叫他钱老师。今年秋天，他七十二岁了。巷子里的老人常说，钱老师这一辈子只做了两件事：下棋[6]，教别人下棋。

## 巷子里的棋馆

2 棋馆在一座两层小楼的一楼，只有两间屋子。屋子里摆着八张旧木桌子，每张桌子上放着一个木棋盘[7]和两个圆圆的棋盒[8]，一个装黑子，一个装白子。墙上挂满了照片，都是这些年在这里学过棋[9]的孩子。有的照片已经发黄了，照片里的孩子，现在大概也有自己的孩子了。

1 杭州 *Hángzhōu* Hangzhou
2 巷子 *xiàngzi* lane, alley
3 西湖 *Xī Hú* West Lake
4 棋馆 *qíguǎn* Go club
5 青石棋馆 *Qīngshí Qíguǎn* Bluestone Go Club
6 下棋 *xiàqí* to play Go
7 棋盘 *qípán* Go board
8 棋盒 *qíhé* bowl for Go stones
9 棋 *qí* Go, board game

Hangzhou has many old lanes. In one small lane not far from West Lake 1
there is a tiny Go club. An old wooden sign hangs by the door, with four characters on it: Bluestone Go Club. The club has been open for more than thirty years. Its owner's surname is Qian, and everyone calls him Teacher Qian. This autumn he turned seventy-two. The old people in the lane often say that Teacher Qian has done only two things in his whole life: play Go, and teach other people to play Go.

## The Go Club in the Lane

The club is on the ground floor of a small two-storey building, and has only 2
two rooms. In the rooms stand eight old wooden tables. On each table is a wooden Go board and two round bowls, one holding the black stones and one the white. The walls are covered with photographs, all of children who have learned Go here over the years. Some of the photos have already turned yellow; the children in them probably have children of their own by now.

2 窗外有一棵老树，夏天的下午，树叶的影子落在棋盘[1]上，跟棋子[2]混在一起，有时候连钱老师也分不清。

3 围棋[3]是世界上最古老的棋[4]之一。据说它已经有四千年的历史了。传说古代有一位叫尧[5]的帝王，他的儿子不爱学习，尧就发明了围棋来教他。这当然只是一个传说，谁也无法证明。不过可以肯定的是，两千多年前的书里就已经有关于下棋[6]的记载了。在古代，读书人要学“琴棋书画[7]”四种本领，其中的“棋”，说的就是围棋。

4 围棋的规则非常简单。棋盘上有十九条横线、十九条直线，一共有三百六十一个交叉点[8]。两个人一个拿黑子，一个拿白子，你一步，我一步，把棋子放在点上，最后谁围住的地方大，谁就赢。

1 棋盘 *qípán* Go board
2 棋子 *qízǐ* Go stone
3 围棋 *wéiqí* Go (board game)
4 棋 *qí* Go, board game
5 尧 *Yáo* Yao, legendary ruler
6 下棋 *xiàqí* to play Go
7 琴棋书画 *qín qí shū huà* zither, Go, calligraphy, painting
8 交叉点 *jiāochādiǎn* intersection

Outside the window stands an old tree. On summer afternoons the shadows 2
of its leaves fall on the boards and mix with the stones, and sometimes even Teacher Qian cannot tell them apart.

Go is one of the oldest board games in the world. It is said to be four thou- 3
sand years old. Legend has it that in ancient times there was a ruler called Yao whose son did not like to study, so Yao invented Go to teach him. Of course this is only a legend, and no one can prove it. What is certain, though, is that books from more than two thousand years ago already contain accounts of the game. In ancient times a scholar was expected to master four arts, "the zither, Go, calligraphy and painting", and the "Go" among them meant exactly this game.

The rules of Go are very simple. The board has nineteen horizontal lines 4
and nineteen vertical ones, making three hundred and sixty-one intersections in all. Two people, one with black stones and one with white, take turns placing stones on the points, and in the end whoever has surrounded more territory wins.

4 钱老师教新学生的第一天，总是说："规则不过是几句话罢了，五分钟就能学会。可是要下好，一辈子也不够。"这句话一点儿也没有说错。有人算过，一盘围棋[1]可能出现的变化，比整个宇宙[2]中原子的数量还要多。也就是说，从古到今，几乎没有两盘完全一样的棋[3]。

5 二十多年前，是青石棋馆[4]最热闹的时候。那时候，每到周末，家长们就带着孩子从城市的各个地方赶来，屋子里坐不下，就在门外的巷子[5]里再摆两张桌子。屋里很安静，只能听见棋子[6]落在木棋盘[7]上的声音，一下，又一下，清清楚楚。钱老师最喜欢这个声音。他说，好的一步棋，声音也是好听的。

6 钱老师年轻的时候在工厂当工人，下班以后才有时间下棋[8]。

1 围棋 *wéiqí* Go (board game)
2 宇宙 *yǔzhòu* universe
3 棋 *qí* Go, board game
4 青石棋馆 *Qīngshí Qíguǎn* Bluestone Go Club
5 巷子 *xiàngzi* lane, alley
6 棋子 *qízǐ* Go stone
7 棋盘 *qípán* Go board
8 下棋 *xiàqí* to play Go

On a new student's first day Teacher Qian always says, "The rules are just a few sentences, that's all; you can learn them in five minutes. But to play well, a lifetime is not enough." He is not wrong at all. Someone has calculated that the number of possible positions in a game of Go is greater than the number of atoms in the whole universe. That is to say, from ancient times until today there have hardly ever been two games exactly alike. 4

Twenty-odd years ago was the club's liveliest time. In those days, every weekend, parents would bring their children from all over the city. When the rooms were full, two more tables were set up in the lane outside the door. Inside it was quiet. All you could hear was the sound of stones landing on the wooden boards, one click, then another, clear and distinct. Teacher Qian loved that sound more than anything. He used to say that a good move sounds good too. 5

As a young man Teacher Qian worked in a factory, and he only had time for Go after his shift. 6

他从来没有当过职业棋手[1]，最好的成绩是全省业余比赛的第三名。四十岁那年，工厂关了门，他用所有的钱租下了这两间屋子。家里人都说他疯了，他却说："我下不过别人，可是我也许能教出下得过别人的人。"

## 二〇一六年

二〇一六年三月，钱老师的生活被一场比赛打乱了。那是一场人和机器之间的比赛：一边是韩国[2]棋手李世石[3]，当时世界上最强的棋手之一；另一边是谷歌[4]公司的一个电脑程序，名字叫"阿尔法狗[5]"。比赛在韩国首尔[6]进行，一共五盘。比赛开始以前，钱老师一点儿也不担心。他对学生们说："电脑只是会计算罢了。围棋[7]可不只是计算。机器想赢李世石，至少还要十年。"

1 棋手 *qíshǒu* chess player
2 韩国 *Hánguó* South Korea
3 李世石 *Lǐ Shìshí* Lee Sedol
4 谷歌 *Gǔgē* Google
5 阿尔法狗 *Ā'ěrfǎgǒu* AlphaGo
6 首尔 *Shǒu'ěr* Seoul
7 围棋 *wéiqí* Go (board game)

He was never a professional; his best result was third place in the provincial amateur championship. The year he turned forty the factory closed, and he used all his money to rent these two rooms. His family said he had gone mad, but he said, "I can't beat other people, but maybe I can teach people who will." 6

**2016**

In March 2016, Teacher Qian's life was turned upside down by a match. It was a match between a human and a machine: on one side the Korean player Lee Sedol, then one of the strongest players in the world; on the other a computer program from Google called AlphaGo. The match was held in Seoul, South Korea, over five games. Before it began, Teacher Qian was not worried at all. He told his students, "A computer can only calculate, that's all. And Go is not just calculation. For a machine to beat Lee Sedol will take at least another ten years." 7

8 第一盘比赛那天，棋馆[1]里挤满了人。大家围在一台小电视前面，看着屏幕上的棋盘[2]。当时十七岁的林远[3]也在。他是钱老师最好的徒弟，也是钱老师最大的骄傲：八岁进棋馆，十六岁就成了职业棋手[4]。那一盘，李世石[5]输了。第二天，他又输了。在第二盘里，阿尔法狗[6]下了一步很奇怪的棋[7]，解说的人都以为是电脑出了错。不料几十步以后，大家才发现，那一步，也就是第三十七步，才是整盘棋的关键。

9 第三盘，李世石又输了。棋馆里没有人说话。到了第四盘，李世石在第七十八步下出了一步谁也没想到的好棋，电脑好像一下子乱了，最后居然认输[8]了。那天晚上，棋馆里的人一起站起来鼓掌，有个老棋迷[9]还哭了。可是第五盘，机器又赢了。四比一。钱老师关了电视，在昏暗的屋子里坐了很久。

1 棋馆 *qíguǎn* Go club
2 棋盘 *qípán* Go board
3 林远 *Lín Yuǎn* Lin Yuan
4 棋手 *qíshǒu* chess player
5 李世石 *Lǐ Shìshí* Lee Sedol
6 阿尔法狗 *Ā'ěrfǎgǒu* AlphaGo
7 棋 *qí* Go, board game
8 认输 *rènshū* to resign
9 棋迷 *qímí* Go fan

On the day of the first game the club was packed. Everyone crowded around a small television, watching the board on the screen. Lin Yuan, seventeen at the time, was there too. He was Teacher Qian's best pupil and his greatest pride: he had come to the club at eight and become a professional player at sixteen. Lee Sedol lost that game. The next day he lost again. In the second game AlphaGo played a very strange move, and the commentators all assumed the computer had made a mistake. But a few dozen moves later, to everyone's surprise, it became clear that this move, move thirty-seven, was the key to the whole game. 8

In the third game Lee Sedol lost again. No one in the club said a word. Then, in the fourth game, with his seventy-eighth move, Lee Sedol played a brilliant move no one had expected. The computer seemed to lose its way all at once, and in the end, amazingly, it resigned. That evening everyone in the club stood up together and applauded, and one old Go fan even cried. But in the fifth game the machine won again. Four to one. Teacher Qian switched off the television and sat in the dim room for a long time. 9

10 一年以后，二〇一七年五月，新版本的阿尔法狗[1]来到了离杭州[2]不远的乌镇[3]，对手是十九岁的中国棋手[4]柯洁[5]。当时，柯洁在世界上排名第一。这一次，人连一盘也没有赢，零比三。据说在最后一盘中，柯洁离开座位很长时间，回来的时候，眼睛是红的。

11 看完最后一盘，林远[6]问钱老师："老师，难道我们下了这么多年的棋[7]，都下错了吗？"钱老师想了很久，才慢慢地说："机器比人强，未必说明人下棋[8]没有意义。汽车比人跑得快，可是人照样跑步。"林远没有说话。那时他还不知道，自己以后每天跟机器在一起的时间，会比跟人在一起的时间还长。

1 阿尔法狗 *Ā'ěrfǎgǒu* AlphaGo
2 杭州 *Hángzhōu* Hangzhou
3 乌镇 *Wūzhèn* Wuzhen, a water town
4 棋手 *qíshǒu* chess player
5 柯洁 *Kē Jié* Ke Jie
6 林远 *Lín Yuǎn* Lin Yuan
7 棋 *qí* Go, board game
8 下棋 *xiàqí* to play Go

A year later, in May 2017, a new version of AlphaGo came to Wuzhen, not far from Hangzhou. Its opponent was the nineteen-year-old Chinese player Ke Jie. At the time Ke Jie was ranked number one in the world. This time the human did not win a single game: zero to three. It is said that during the last game Ke Jie left his seat for a long time, and when he came back, his eyes were red. 10

After watching the last game, Lin Yuan asked Teacher Qian, "Teacher, are you telling me that all these years we've been playing Go wrong?" Teacher Qian thought for a long time before answering slowly, "A machine being stronger than people doesn't necessarily mean it's pointless for people to play. Cars are faster than people, but people go running all the same." Lin Yuan said nothing. He did not yet know that one day he would spend more time each day with machines than with people. 11

# 走了的学生

12 那两场比赛以后，围棋[†]的世界很快就变了。不到两年，比人还强的围棋程序就出现在了每个人的手机上，而且很多是免费的。孩子们在家里就能跟“AI老师”下棋[‡]。下完以后，程序会告诉你每一步是好是坏，还会给出一个数字——胜率[§]。胜率，就是说，下了这一步以后，你赢这盘棋[¶]的可能性还有百分之几。二〇一九年，李世石[*]宣布退役[#]。他说，就算自己成为第一，也还有一个打不败的对手在那儿。

13 一开始，钱老师觉得这些程序只是玩具。后来，学生一个一个地少了。有的家长很客气地对他说：“钱老师，孩子学习忙，下学期就不来了。”有的说得很直接：“手机上的老师不会累，二十四小时都能上课，也不要钱。”钱老师笑着点头，说：“好，好，有空回来玩。”

† 围棋 *wéiqí* Go (board game)
‡ 下棋 *xiàqí* to play Go
§ 胜率 *shènglǜ* winning probability
¶ 棋 *qí* Go, board game
* 李世石 *Lǐ Shìshí* Lee Sedol
# 退役 *tuìyì* to retire (from competition)

## The Students Who Left

After those two matches, the world of Go changed quickly. In less than two years, Go programs stronger than any human had appeared on everyone's phone, and many of them were free. Children could play against an "AI teacher" without leaving home. After a game, the program would tell you whether each move was good or bad, and it would also give a number: the win rate. The win rate means, in other words, how many percent chance you still have of winning the game after making that move. In 2019, Lee Sedol announced his retirement. He said that even if he became number one, there would still be an opponent out there that could not be beaten. 12

At first, Teacher Qian thought these programs were only toys. Then, one by one, his students dropped away. Some parents told him politely, "Teacher Qian, our child is busy with schoolwork, so he won't be coming next term." Others put it bluntly: "The teacher on the phone never gets tired, can give lessons twenty-four hours a day, and costs nothing." Teacher Qian would nod with a smile and say, "Fine, fine. Come back and play when you have time." 13

13 可是走了的孩子，很少有回来的。

14 最多的时候，棋馆[†]有六十多个学生，到了去年，只剩下七个。至于那些常来下棋[‡]的老朋友，有的搬到了儿女家，有的身体不好，再也来不了了，还有几个，现在也在手机上下棋。他们不好意思告诉钱老师，可钱老师都知道。巷子[§]里路过的人往里看一眼，常常只看见一个白头发的老人，一个人坐在桌子前面，自己跟自己下棋。

15 只有一个叫小满[¶]的女孩，每个星期六下午都来。她十岁，下棋很快，说话也很快。有一次，她下完一盘，拿出手机看了看，认真地说：“钱老师，AI说您第三十二步下错了，胜率[*]掉了百分之八。”钱老师哈哈大笑：“它说得对，我是故意下错的。”小满不明白：“故意？为什么？”“因为你一直在进攻，我想看看你会不会防守。”

† 棋馆 *qíguǎn* Go club
‡ 下棋 *xiàqí* to play Go
§ 巷子 *xiàngzi* lane, alley
¶ 小满 *Xiǎomǎn* Xiaoman
* 胜率 *shènglǜ* winning probability

But few of the children who left ever came back. 13

At its busiest the club had more than sixty students; by last year only seven were left. As for the old friends who used to come and play, some had moved in with their children, some were in poor health and could no longer come, and a few now played on their phones too. They were too embarrassed to tell Teacher Qian, but he knew all the same. People passing in the lane who glanced inside would often see only a white-haired old man sitting alone at a table, playing Go against himself. 14

Only one girl, called Xiaoman, still came every Saturday afternoon. She was ten, played fast and talked fast. Once, after finishing a game, she took out her phone, looked at it and said seriously, "Teacher Qian, the AI says your thirty-second move was a mistake. Your win rate dropped eight percent." Teacher Qian burst out laughing. "It's right. I made that mistake on purpose." Xiaoman didn't understand. "On purpose? Why?" "Because you were attacking the whole time, and I wanted to see whether you knew how to defend." 15

小满[†]一时没明白，想了想，才说：“AI可不会这样。”钱老师说：“对，所以它只是一个对手罢了，不是老师。”

今年夏天，房东来了。他说，这一带要改造，房子要重新装修，明年房租要涨一倍。这几年，棋馆[‡]只靠几个学生的学费支撑着。就是说，棋馆一年的收入，连半年的房租也交不起。房东走了以后，钱老师在门口坐了一个下午，看着巷子[§]里的人来来往往，不时叹气。他想，也许是时候了。

## 回来的人

十月的一个傍晚，下着小雨。屋子里已经有些昏暗，钱老师正在收拾棋盒[¶]，门口忽然有人叫了一声：“老师。”他抬起头，看见一个高高的年轻人站在雨里，手里拿着一把黑伞。

† 小满 *Xiǎomǎn* Xiaoman
‡ 棋馆 *qíguǎn* Go club
§ 巷子 *xiàngzi* lane, alley
¶ 棋盒 *qíhé* bowl for Go stones

For a moment Xiaoman didn't follow. She thought about it, then said, "The AI would never do that." "Exactly," said Teacher Qian. "That's why it's just an opponent, nothing more. It isn't a teacher." 15

This summer, the landlord came. He said the whole area was going to be redeveloped and the building renovated, and next year the rent would double. For years the club had been kept going only by a few students' fees. In other words, a whole year's income from the club would not even cover six months' rent. After the landlord left, Teacher Qian sat by the door all afternoon, watching people come and go in the lane and sighing now and then. Perhaps, he thought, it was time. 16

## The One Who Came Back

One evening in October, a light rain was falling. The room was already growing dim. Teacher Qian was tidying away the stone bowls when someone at the door suddenly called out, "Teacher." He looked up and saw a tall young man standing in the rain, holding a black umbrella. 17

17 他看了好一会儿，才认出是自己的老徒弟，笑了：“林远[†]！”十年过去了，林远已经二十七岁，是职业七段[‡]，平时住在北京，很少回杭州[§]。钱老师给他泡了一杯龙井[¶]，两个人坐在最里面那张桌子旁边，那是林远小时候常坐的位置。

18 林远看了看空空的屋子，问：“老师，听说房租要涨了？”钱老师说：“消息倒是传得快。”林远放下茶杯，好像早就想好了要说的话：“老师，我觉得棋馆[*]……要么关了，要么搬到网上去。”“现在的孩子都在网上学棋[#]。您在网上开课，一个晚上能教几百个学生，还能用AI帮他们分析。我有朋友做这个，我可以帮您。”

19 钱老师听完，摇了摇头：“我不过是个下了一辈子棋的老头子罢了。对着电脑说话，我说不出来。”

† 林远 *Lín Yuǎn* Lin Yuan
‡ 七段 *qīduàn* seventh dan
§ 杭州 *Hángzhōu* Hangzhou
¶ 龙井 *lóngjǐng* Longjing tea
* 棋馆 *qíguǎn* Go club
\# 棋 *qí* Go, board game

He looked for quite a while before he recognised his old pupil, and smiled. 17
"Lin Yuan!" Ten years had passed. Lin Yuan was now twenty-seven, a professional seventh dan. He lived in Beijing and seldom came back to Hangzhou. Teacher Qian made him a cup of Longjing tea, and the two of them sat down at the table at the very back, the place where Lin Yuan used to sit as a boy.

Lin Yuan looked around the empty rooms and asked, "Teacher, I hear the 18
rent is going up?" "News certainly travels fast," said Teacher Qian. Lin Yuan put down his cup, as if he had long ago worked out what he wanted to say. "Teacher, I think the club should... either close, or move online." "Kids today all learn Go online. If you taught classes online, you could teach hundreds of students in one evening, and use AI to help analyse their games. I have friends who do this; I could help you."

Teacher Qian heard him out and shook his head. "I'm just an old man who 19
has played Go all his life, that's all. I couldn't talk to a computer; the words wouldn't come."

19 林远[†]说：“至于技术上的事，您完全不用担心，都交给我们。难道您要一个人守着这间空屋子吗？”钱老师没有回答，却问他：“你呢？你现在每天怎么练棋[‡]？”林远沉默了一会儿，说：“每天八个小时，跟AI下，研究AI的棋。老师，说实话，现在职业比赛里的很多步，都是AI教的。不这样练，就会输。”

20 他又说，这几年他的成绩其实不错，可是他越来越不喜欢下棋[§]了。“以前我输了，会一个人想一晚上，为什么会输。现在输了，打开AI看一眼就知道了：第几步错了，胜率[¶]掉了多少。”他笑了笑，说：“很快，也很清楚，可是好像跟我没有什么关系。赢的时候也一样，我常常不知道，赢的是我，还是我背下来的那些AI的棋。我下的棋，未必是我自己的棋。”

† 林远 *Lín Yuǎn* Lin Yuan
‡ 棋 *qí* Go, board game
§ 下棋 *xiàqí* to play Go
¶ 胜率 *shènglǜ* winning probability

"As for the technical side, you don't need to worry about that at all; leave it all to us," said Lin Yuan. "Do you really want to sit alone in this empty room?" Teacher Qian didn't answer. Instead he asked, "And you? How do you train these days?" Lin Yuan was silent for a moment. "Eight hours a day. I play against AI and study its games. To be honest, Teacher, a lot of the moves in professional games now were taught by AI. If you don't train like this, you lose." 19

He went on to say that his results these past few years had actually been good, but he liked Go less and less. "When I lost, I used to spend the whole night alone working out why. Now when I lose, I open the AI, take one look and I know: which move was wrong and how much my win rate dropped." He gave a little smile. "It's fast and it's clear, but somehow it has nothing to do with me. Winning is the same. I often can't tell whether it's me who has won, or the AI moves I learned by heart. The games I play aren't necessarily my own." 20

21 钱老师看着他，看了很久。窗外的雨越下越大。他把桌子上的棋盘[1]擦了擦，把一盒黑子拉到自己面前，把白子推给了林远[2]。“那我们下一盘吧。”他说，“最后一盘。你赢了，棋馆[3]就关，我听你的。”林远想说什么，最后还是没有说出口。他慢慢地打开了棋盒[4]。

## 最后一局

22 那天晚上，小满[5]也来了。她本来是来还一本棋书[6]的，看到两个人坐在棋盘两边，就安安静静地坐在了旁边。钱老师拿黑子，先下。开局[7]没多久，林远就下了一步让钱老师很吃惊的棋[8]：他把一颗白子直接放进了黑棋[9]的角里。二十年前，这样的下法是要被老师骂的，因为大家都认为它太早、太急。

1 棋盘 *qípán* Go board
2 林远 *Lín Yuǎn* Lin Yuan
3 棋馆 *qíguǎn* Go club
4 棋盒 *qíhé* bowl for Go stones
5 小满 *Xiǎomǎn* Xiaoman
6 棋书 *qíshū* book on Go
7 开局 *kāijú* opening (of a game)
8 棋 *qí* Go, board game
9 黑棋 *hēiqí* black's stones

Teacher Qian looked at him for a long time. Outside the window the rain 21
grew heavier. He wiped the board on the table, pulled a bowl of black stones towards himself and pushed the white ones over to Lin Yuan. "Then let's play a game," he said. "One last game. If you win, the club closes. I'll do as you say." Lin Yuan wanted to say something, but in the end the words would not come out. Slowly, he opened the bowl.

## The Last Game

That evening, Xiaoman came too. She had only come to return a book on 22
Go, but when she saw the two of them sitting on either side of the board, she sat down quietly beside them. Teacher Qian took black and moved first. Not long into the opening, Lin Yuan played a move that startled Teacher Qian: he placed a white stone straight into black's corner. Twenty years ago a teacher would have scolded you for playing like that, because everyone thought it was too early and too hasty.

22 可是自从AI这样下过以后，几乎所有的职业棋手[1]都学会了。

23 “这也是机器教的？”钱老师问。“是。”林远[2]说，“AI算过，这样下胜率[3]更高。”钱老师没有再问，只是点了点头。接下来的几十步，林远全神贯注[4]，下得又快又准，好像每一步都不用想。钱老师却下得很慢，时而看看棋盘[5]，时而看看窗外的雨，有时候拿起一颗棋子[6]，在手里转了半天，才轻轻放下去。小满[7]看看老师，又看看林远，觉得他们两个好像不是在下同一盘棋[8]。

24 下到第四十一步，钱老师把一颗黑子放在了棋盘左边一个谁也想不到的地方。这步棋看起来很慢，几乎没有用处，还让白棋[9]多得了一块地。林远看了一眼，心里想：老师真的老了，这样的错误，他以前是不会犯的。

1 棋手 *qíshǒu* chess player
2 林远 *Lín Yuǎn* Lin Yuan
3 胜率 *shènglǜ* winning probability
4 全神贯注 *quánshén guànzhù* totally absorbed
5 棋盘 *qípán* Go board
6 棋子 *qízǐ* Go stone
7 小满 *Xiǎomǎn* Xiaoman
8 棋 *qí* Go, board game
9 白棋 *báiqí* white's stones

But ever since AI started playing this way, almost every professional has learned it. 22

"Did the machine teach you this too?" asked Teacher Qian. "Yes," said Lin Yuan. "The AI has worked it out. Playing like this gives a higher win rate." Teacher Qian asked nothing more. He only nodded. For the next few dozen moves Lin Yuan was totally absorbed, playing fast and precisely, as if he didn't need to think about any of them. Teacher Qian, though, played slowly, now looking at the board, now at the rain outside the window. Sometimes he would pick up a stone and turn it over in his fingers for ages before gently putting it down. Xiaoman looked at the teacher, then at Lin Yuan, and felt as if the two of them weren't playing the same game. 23

On the forty-first move, Teacher Qian placed a black stone at a point on the left side of the board that no one would have thought of. It looked very slow, almost useless, and it even let white gain a piece of territory. Lin Yuan glanced at it and thought: the old man really has grown old. He would never have made a mistake like that before. 24

24 他没有多想，按照AI的想法，抢先占了右边最大的地方。小满[1]在旁边小声说：“这步棋[2]，AI一定不喜欢。”钱老师笑了笑，没有说话。

25 可是又下了二十多步以后，林远[3]发现事情没有那么简单。左边那颗看起来没有用的黑子，居然慢慢地跟别的黑子连了起来，变成了一道墙。他的一大块白棋[4]被围在里面，要花很多步才能救出来。他停了下来，全神贯注[5]地看着棋盘[6]，整整想了十分钟。在职业比赛里，他已经很久没有为一步棋想这么久了。他忽然觉得，对面坐着的不是一个七十多岁的老人，而是一个他不认识的对手。

26 最后，林远还是赢了。他毕竟是每天跟AI训练的职业棋手[7]，后半盘几乎没有出错。数完棋子[8]，白棋赢了三目半[9]。

1 小满 *Xiǎomǎn* Xiaoman
2 棋 *qí* Go, board game
3 林远 *Lín Yuǎn* Lin Yuan
4 白棋 *báiqí* white's stones
5 全神贯注 *quánshén guànzhù* totally absorbed
6 棋盘 *qípán* Go board
7 棋手 *qíshǒu* chess player
8 棋子 *qízǐ* Go stone
9 目 *mù* point (of territory in Go)

Without thinking further, he did what the AI would do and took the biggest area on the right first. Beside them, Xiaoman whispered, "The AI definitely wouldn't like that move." Teacher Qian smiled and said nothing. 24

But twenty-odd moves later, Lin Yuan found that things were not so simple. That apparently useless black stone on the left had, to his surprise, slowly joined up with the other black stones and become a wall. A large group of his white stones was shut in behind it, and it would take many moves to rescue them. He stopped, his eyes fixed on the board, and thought for a full ten minutes. It had been a long time since he had spent so long over a single move, even in a professional game. He suddenly felt that the person sitting opposite him was not a man in his seventies but an opponent he did not know. 25

In the end, Lin Yuan still won. He was, after all, a professional who trained with AI every day, and he made almost no mistakes in the second half of the game. When the stones were counted, white had won by three and a half points. 26

26 在围棋[1]里，这是很小的差距，就是说，只要有一两步下得不一样，赢的人也许就是黑棋[2]了。钱老师笑了笑，把手里最后一颗黑子放回棋盒[3]："你赢了。棋馆[4]关门。"林远[5]却没有笑，若有所思[6]。他看着棋盘[7]左边那一片黑子，总觉得它们好像在跟他说什么。

## 旧本子

27 钱老师没有马上收棋子[8]。他站起来，走进里屋，拿出一个旧铁盒子。盒子里是十几个发黄的本子。"你还记得这些吗？"他翻开其中一本，第一页上写着：林远，八岁。下面是一盘棋[9]的记录，每一步都用小字写得清清楚楚，旁边还有红笔写的几句话。

1 围棋 *wéiqí* Go (board game)
2 黑棋 *hēiqí* black's stones
3 棋盒 *qíhé* bowl for Go stones
4 棋馆 *qíguǎn* Go club
5 林远 *Lín Yuǎn* Lin Yuan
6 若有所思 *ruòyǒusuǒsī* lost in thought
7 棋盘 *qípán* Go board
8 棋子 *qízǐ* Go stone
9 棋 *qí* Go, board game

In Go this is a very small margin. In other words, if just one or two moves had 26
been different, the winner might well have been black. Teacher Qian smiled and put the last black stone in his hand back into the bowl. "You win. The club closes." But Lin Yuan did not smile; he seemed lost in thought. He looked at the group of black stones on the left of the board and could not shake the feeling that they were trying to tell him something.

## The Old Notebooks

Teacher Qian did not clear away the stones right away. He stood up, went 27
into the back room and came out with an old tin box. Inside were a dozen or so yellowed notebooks. "Do you still remember these?" He opened one of them. On the first page was written: Lin Yuan, age eight. Below it was the record of a game, every move written out clearly in small characters, with a few lines in red pen beside it.

27 “这是你在棋馆[†]下的第一盘棋[‡]。”钱老师说，“你在这里下的每一盘棋，我都记下来了。”林远[§]拿着本子，一时说不出话来。

28 钱老师照着本子，把那盘旧棋在旁边的棋盘[¶]上重新摆了一遍。摆到第三十多步的时候，林远一下子坐直了。那个八岁的孩子，也在左边下过一步一模一样的棋：一颗看起来没有用、当时谁都看不懂的黑子。钱老师今天下的，恰恰就是这一步。“我那时候……为什么下在这里？”他问。钱老师指着本子上的红字。那是他十九年前写的：“问他为什么下这里，他说：我想给后面的棋留一条路。这孩子，将来会下得很好。”

† 棋馆 *qíguǎn* Go club
‡ 棋 *qí* Go, board game
§ 林远 *Lín Yuǎn* Lin Yuan
¶ 棋盘 *qípán* Go board

"This is the first game you played at the club," said Teacher Qian. "I wrote down every game you ever played here." Lin Yuan held the notebook and for a moment could not say a word. 27

Following the notebook, Teacher Qian laid out that old game again on the board next to them. When he reached the thirty-somethingth move, Lin Yuan suddenly sat up straight. That eight-year-old had also played exactly the same move on the left: a black stone that looked useless, one that nobody at the time could make sense of. And what Teacher Qian had played today was precisely that move. "Why did I... play here back then?" he asked. Teacher Qian pointed to the red writing in the notebook. He had written it nineteen years earlier: "Asked him why he played here. He said: I want to leave a road open for the stones that come later. This child will play very well one day." 28

29 “AI会告诉你，这一步胜率[1]掉了百分之十。”钱老师说，“它说得未必不对。可是，恰恰是这样的‘坏棋[2]’，让你今天想了十分钟。它不会问你为什么这样下，也不会记得一个八岁的孩子曾经这样想过。”他停了一下，又说：“古时候的人把下围棋[3]叫作‘手谈[4]’，就是说，两个人不说话，用手在棋盘[5]上谈话。”“今天这一盘，我一直在跟你说话。可是你呢？你一直在跟机器说话。”

30 林远[6]低着头，很久没有说话。过了一会儿，他小声说：“老师，可是不跟机器学，我就会输。”“那就学。”钱老师说，“机器比我们强，这是事实，用不着遗憾，也用不着害怕。它能教你怎么赢，这很好。”“至于你为什么下棋[7]，下棋的时候心里想的是什么，这个问题它回答不了，只有你自己能回答。”

1 胜率 *shènglǜ* winning probability
2 棋 *qí* Go, board game
3 围棋 *wéiqí* Go (board game)
4 手谈 *shǒután* "hand talk", old name for Go
5 棋盘 *qípán* Go board
6 林远 *Lín Yuǎn* Lin Yuan
7 下棋 *xiàqí* to play Go

"The AI will tell you that this move drops your win rate by ten percent," 29
said Teacher Qian. "It may well be right. Yet it was precisely this kind of 'bad move' that made you think for ten minutes today. It will never ask you why you played it, and it won't remember that an eight-year-old once thought this way." He paused, then went on. "In the old days people called playing Go 'hand talk'. That is to say, two people talk on the board with their hands, without saying a word." "All through tonight's game I was talking to you. But you? You were talking to a machine the whole time."

Lin Yuan kept his head down and said nothing for a long time. After a while 30
he said quietly, "But Teacher, if I don't learn from the machines, I'll lose." "Then learn," said Teacher Qian. "Machines are stronger than we are. That's a fact, and there's no need to regret it, or to be afraid. It can teach you how to win, and that's good." "But as for why you play, and what is in your heart when you play, that's a question it can't answer. Only you can."

30 他把本子放回铁盒子，推到林远[1]面前：“这些给你。棋馆[2]关了，它们放在这儿也没用。”

## 棋子[3]的声音

31 那一年冬天，青石棋馆[4]还是关了门。林远说过要帮老师交房租，钱老师没有答应。林远觉得很遗憾，钱老师自己却一点儿也不遗憾。“棋馆不过是一间房子罢了，”他说，“棋[5]又不在房子里。”他把八张桌子送给了附近的一所小学，只留下两个最旧的木棋盘[6]。每个星期六下午，他在社区的活动室里教孩子们下棋[7]，不收钱。第一个星期，只来了小满[8]一个人。第二个星期，她带来了两个同学。到了第二年春天，活动室里已经坐不下了。

1 林远 *Lín Yuǎn* Lin Yuan
2 棋馆 *qíguǎn* Go club
3 棋子 *qízǐ* Go stone
4 青石棋馆 *Qīngshí Qíguǎn* Bluestone Go Club
5 棋 *qí* Go, board game
6 棋盘 *qípán* Go board
7 下棋 *xiàqí* to play Go
8 小满 *Xiǎomǎn* Xiaoman

He put the notebooks back into the tin box and pushed it across to Lin Yuan. 30
"These are for you. Once the club is closed, they'll be no use sitting here."

## The Sound of the Stones

That winter, the Bluestone Go Club closed its doors after all. Lin Yuan had 31
offered to pay the rent for him, but Teacher Qian refused. Lin Yuan was sorry about it; Teacher Qian himself had no regrets at all. "A club is only a room, that's all," he said. "Go doesn't live in a room." He gave the eight tables to a primary school nearby, and kept only the two oldest wooden boards. Every Saturday afternoon he taught children Go in the neighbourhood activity room, free of charge. The first week, only Xiaoman came. The second week, she brought two classmates. By the following spring there was no longer room for everyone in the activity room.

32 那年春天，林远[1]在一场重要的比赛里，下出了一步很奇怪的棋[2]。网上直播的解说员说：“这步棋，AI显示的胜率[3]很低，不知道林[4]七段[5]在想什么。”至于那盘棋林远最后是赢了还是输了，钱老师并不太关心。可是他心里有一点骄傲。他只记得比赛结束以后，有记者问林远为什么那样下。林远若有所思[6]地想了想，说：“这步棋不是AI教我的。是我八岁的时候，自己想出来的。”

33 那天下午，活动室里很安静，只能听见棋子[7]落在木棋盘[8]上的声音，一下，又一下。小满[9]忽然抬起头，问：“钱老师，在手机上下棋[10]，也有这种声音吗？”钱老师想了想，说：“有，不过是电脑做出来的。”“那有什么不一样？”钱老师拿起一颗黑子，轻轻地放在棋盘上。“啪”[11]的一声，很轻，很清楚。

1 林远 *Lín Yuǎn* Lin Yuan
2 棋 *qí* Go, board game
3 胜率 *shènglǜ* winning probability
4 林 *Lín* Lin (surname)
5 七段 *qīduàn* seventh dan
6 若有所思 *ruòyǒusuǒsī* lost in thought
7 棋子 *qízǐ* Go stone
8 棋盘 *qípán* Go board
9 小满 *Xiǎomǎn* Xiaoman
10 下棋 *xiàqí* to play Go
11 啪 *pā* click, snap (sound)

That spring, in an important match, Lin Yuan played a very strange move. 32
The commentator on the online broadcast said, "The AI shows a very low win rate for this move. I have no idea what Lin Seventh Dan is thinking." As for whether Lin Yuan won or lost that game in the end, Teacher Qian did not much care. But he felt a little proud. All he remembered was that after the game a reporter asked Lin Yuan why he had played that way. Lin Yuan thought it over for a moment and said, "The AI didn't teach me this move. I thought of it myself, when I was eight."

That afternoon the activity room was quiet. All you could hear was the 33
sound of stones landing on the wooden boards, one click, then another. Suddenly Xiaoman looked up and asked, "Teacher Qian, when you play on a phone, is there a sound like this too?" Teacher Qian thought for a moment. "There is, but the computer makes it." "So what's the difference?" Teacher Qian picked up a black stone and set it gently on the board. Click: soft, and very clear.

33 “你听，”他说，“这一声，是你自己放下去的。”

"Listen," he said. "That sound is one you put down yourself." 33